

3 (**short**) A cylindrical container with radius R and height H rotates at angular rate Ω as sketched in Fig. 3. Within the container is an incompressible liquid; gravity is downwards as shown.

(a) Derive an expression for the radial variation of liquid depth, h , in terms of the depth at the centre-line h_o , local radius r , Ω and g . [4]

(b) If the total liquid volume is Q , find the depth at the centre-line h_o in terms of Q , R , g and Ω . [2]

(c) If the liquid depth at radius R is H derive an expression for the volume Q in terms of h_o , H and R . [4]

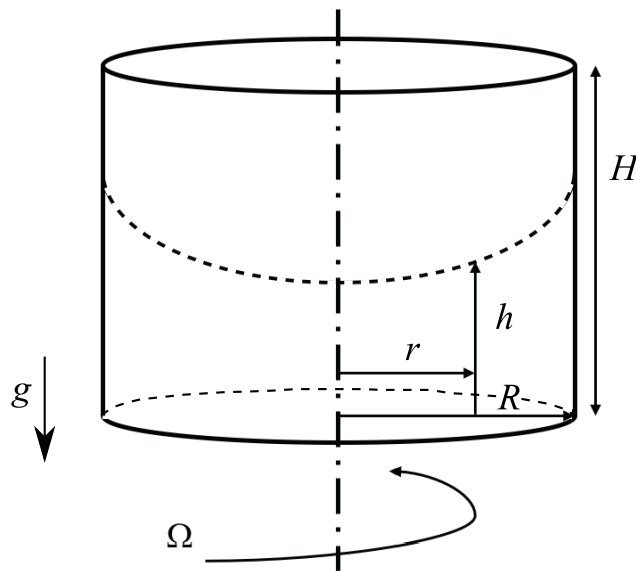


Fig. 3